

Data Preprocessing

Data exploration, cleaning, transformation, splitting and dimensionality reduction. Student Graduation is the main dataset. Food Bank crop production supplements tasks 2 and 3 with actual imputation and transformation (Section 7). Both analyses use the supplied CSV files.

1. Data Exploration

1a. Structure, summary statistics and data types

Measure	Result
Dataset size	4,424 rows; 35 columns
Predictors / target	34 predictors / Target
Feature groups	17 numerical and 17 categorical predictors
Stored data types	29 integer, 5 floating-point and 1 text column
Missing values / duplicate rows	0 / 0

First five rows: selected columns

Rows 1-5 are the first five records in file order, not a random sample. This excerpt shows 6 of the 35 columns; the notebook displays all columns. Category codes are preserved as supplied.

Row	Course	Gender	Age	CU1 grade	CU2 grade	Target
1	2	1	20	0.00	0.00	Dropout
2	11	1	19	14.00	13.67	Graduate
3	5	1	19	0.00	0.00	Dropout
4	15	0	20	13.43	12.40	Graduate
5	3	0	45	12.33	13.00	Graduate

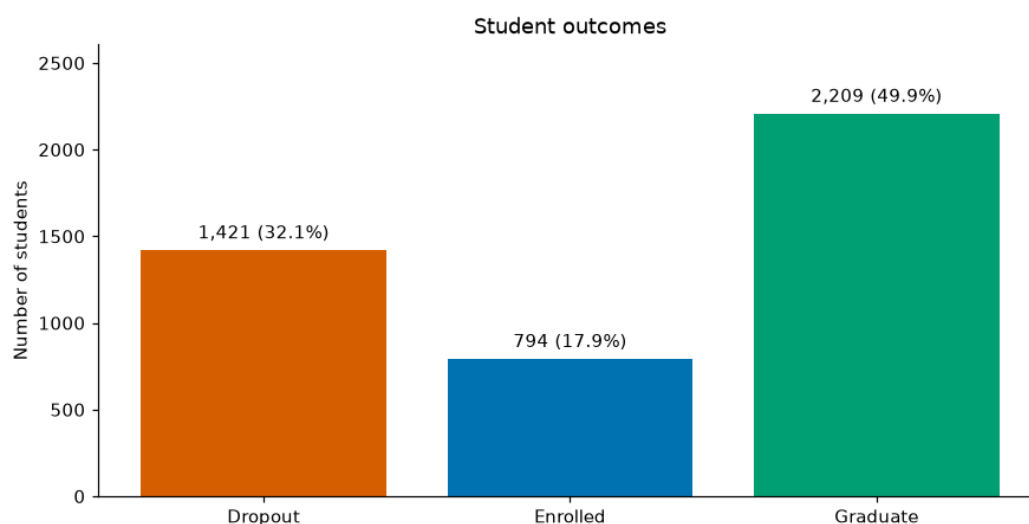


Figure 1. Outcome counts and percentages across all 4,424 students.

Appendix A lists every column and its stored datatype. Appendix B reports summary statistics for all numerical features. The notebook contains the complete `head()` and `describe()` outputs.

Figure 1 shows that Graduate is the largest class (49.9%) and Enrolled the smallest (17.9%). This imbalance motivates a stratified split. A classifier that always predicts Graduate would achieve about 50% accuracy; later evaluation should therefore also examine performance per class.

1. Data Exploration (continued)

1b. Missing values and categorical values

No column contains missing values. Most categorical predictors are stored as integers, but these codes represent categories rather than measured quantities. The numerical features are assessed for potential outliers in Section 3.

Categorical feature	Unique	Observed values
Marital status	6	1, 2, 3, 4, 5, 6
Application mode	18	1-18 (all integers)
Course	17	1-17 (all integers)
Daytime/evening attendance	2	0, 1
Previous qualification	17	1-17 (all integers)
Nationality	21	1-21 (all integers)
Mother's qualification	29	1-29 (all integers)
Father's qualification	34	1-34 (all integers)
Mother's occupation	32	1-32 (all integers)
Father's occupation	46	1-46 (all integers)
Displaced	2	0, 1
Educational special needs	2	0, 1
Debtor	2	0, 1
Tuition fees up to date	2	0, 1
Gender	2	0, 1
Scholarship holder	2	0, 1
International	2	0, 1
Target	3	Dropout, Enrolled, Graduate

2. Data Cleaning

Handling missing values

The dataset contains zero missing values, so no imputation, row deletion or column deletion is required. All 4,424 observations are retained. Introducing replacement values when nothing is missing would unnecessarily change the data.

Numeric zeros are not interpreted as missing: they can be valid observations, such as zero approved curricular units. Missing-value treatment should be based on the meaning of the variables, rather than replacing every zero automatically.

3. Handling Outliers

3a. Detection using the IQR rule

Values below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ are flagged, where $IQR = Q3 - Q1$. This quartile-based rule is less sensitive to extreme values than a mean-and-standard-deviation threshold and does not require approximate normality. It is used for inspection, not automatic deletion.

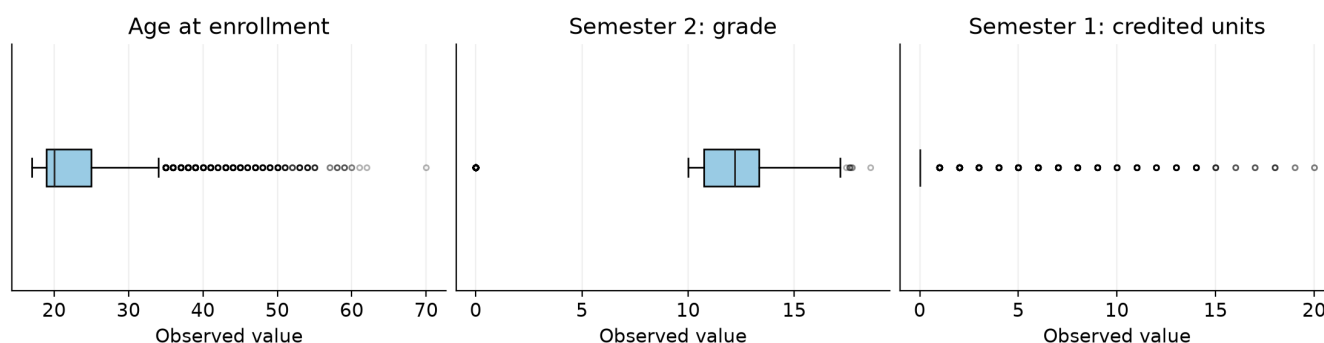


Figure 2. Representative numerical features, each on its own scale. Boxes show the middle 50%; points beyond the $1.5 \times IQR$ whiskers are potential outliers.

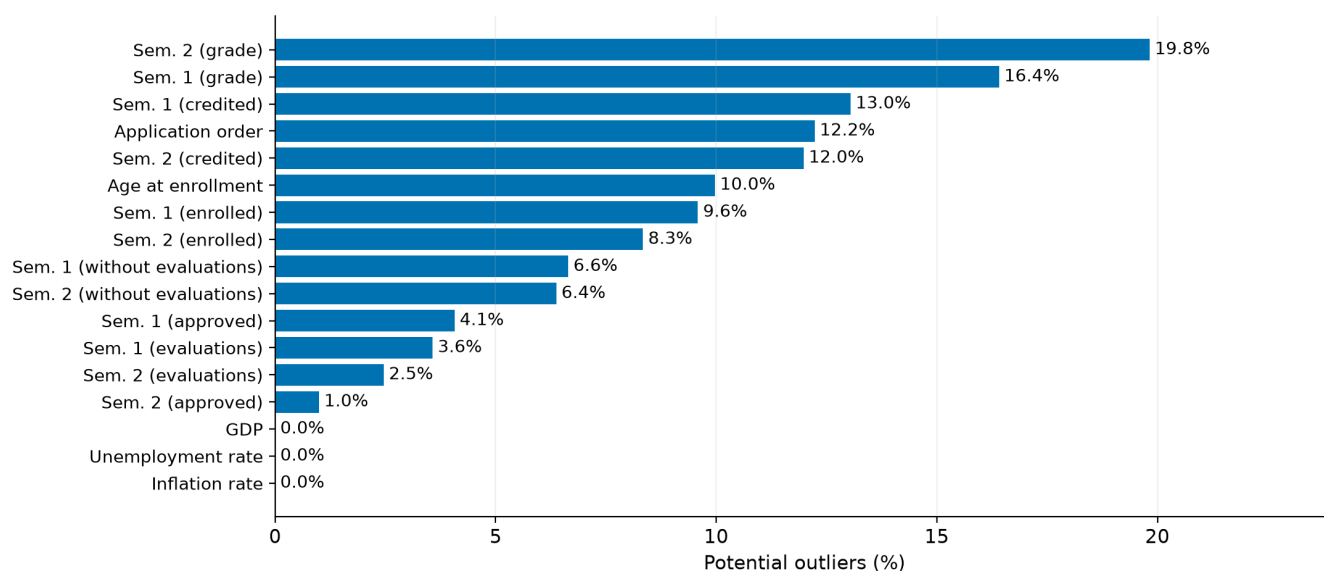


Figure 3. Share of observations flagged in each numerical feature. The same student may be flagged in multiple features; these percentages must not be added together.

3b. Decision and justification

Figure 2 shows a right-skewed age distribution (median 20; middle 50%: 19-25). The upper IQR bound is 34, so 441 older students (10.0%) are flagged. Ages up to 70 are plausible, and rarity alone does not justify deletion. Figure 3 shows the highest flag rate for second-semester grades (19.8%); Figure 2 places zero grades below the typical range.

Credited units and units without evaluations have $Q1 = Q3 = 0$ in both semesters, so all positive values are flagged. The rule therefore marks rarity rather than demonstrating errors. Checks found no approved-unit count above evaluations and no positive approved-unit count with a zero grade. Maximum application order 9, grade 18.875 and evaluations 45 also revealed no clear error. All observations are retained: deletion or capping lacks sufficient justification. These limited checks do not prove that every value is correct.

4. Data Transformation

4a. Encoding categorical data

Feature type	Treatment	Reason
9 multi-category predictors	One-hot encoding	Codes are labels, not ordered quantities.
8 binary predictors	Keep as 0/1	Already have a numerical representation.
Target	Label mapping	Dropout=0, Enrolled=1, Graduate=2.

One-hot encoding expands the 34 predictors to 245 features without implying distances between category codes. The target mapping identifies classes and does not express an ordering. `pandas.get_dummies` is applied to the fixed dataset; for a deployment pipeline, the encoder should be fitted on training data and handle unseen categories.

4b. Feature scaling

The 17 numerical features are standardized to training-set mean zero and standard deviation one. Binary and one-hot indicators remain 0/1. This prevents large numerical units alone from dominating PCA or distance-based models. Figure 4 shows that scaling preserves skewness and potential outliers. Section 5 explains how the scaler is fitted without using test data.

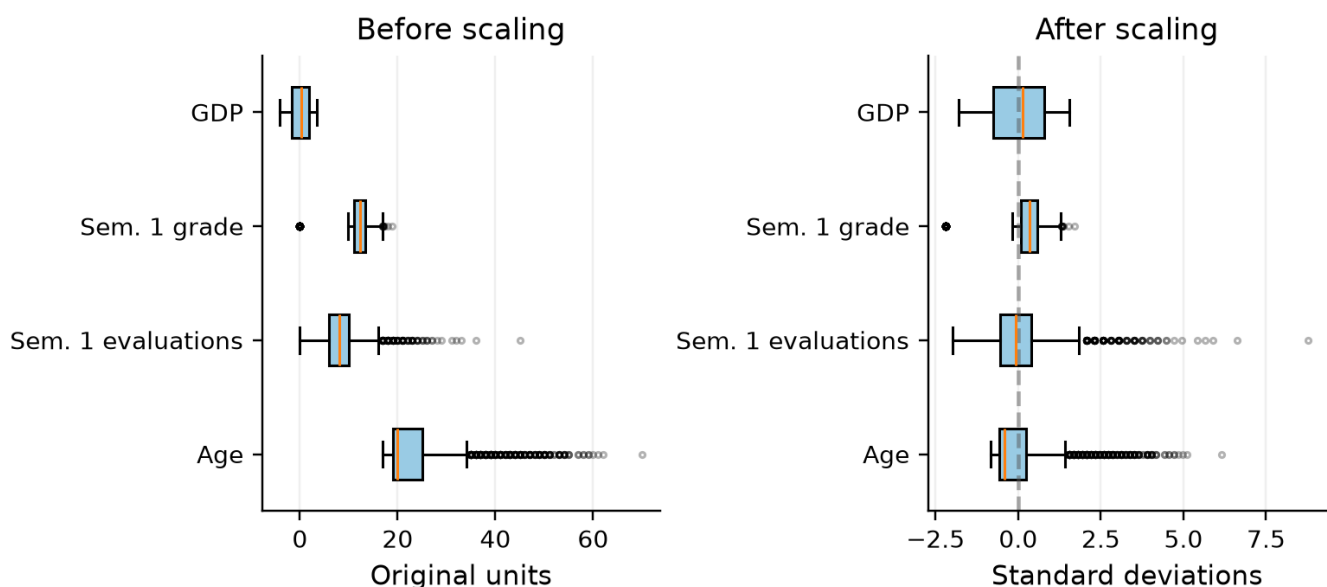


Figure 4. Training features in their original units and in standard deviations from their training mean. Sem. 1 denotes the first semester.

5. Data Splitting

5a. Training and testing sets

The 80/20 split produces **3,539 training students** and **885 test students**.

The split uses `test_size=0.2`, `random_state=42` and `stratify=y`. Class shares are preserved to rounding: Graduate 49.9% in both sets, Dropout 32.1% in both, and Enrolled 17.9% in training versus 18.0% in testing. Splitting occurs before scaling, despite the report section order.

5b. Importance of splitting

The model learns from training observations, while the test set is reserved for final evaluation on unseen students. This helps detect overfitting and avoids evaluating on the same records used to learn. Splitting does not itself prevent overfitting. The scaler and PCA are fitted only on training data and then applied unchanged to test data; their learned parameters therefore do not use the held-out set.

6. Principal Component Analysis (Bonus)

PCA uses `n_components=0.95` and `svd_solver="full"` to retain at least 95% of training variance. A full PCA fit on the same training matrix supplies the complete curve in Figure 5. The fitting procedure follows Section 5.

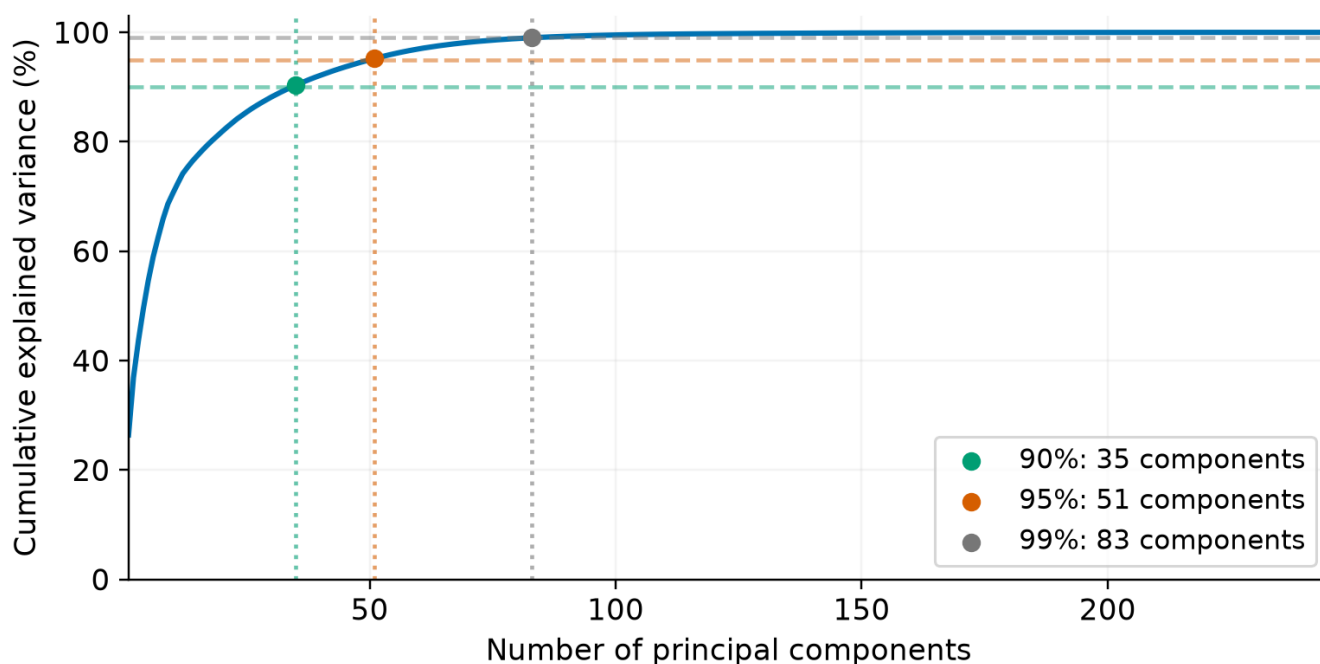


Figure 5. Cumulative training variance retained as components are added. Components are combinations of input features, not a selected subset of original features.

Effect on the dataset

The selected PCA reduces 245 encoded features to 51 components, a 79.18% reduction in dimensions, while retaining 95.22% of variance. Training shape changes from 3,539 x 245 to 3,539 x 51; test shape changes from 885 x 245 to 885 x 51. Figure 5 flattens as more components are added: reaching 99% requires 83 components.

This compression is consistent with overlapping features. For example, first- and second-semester enrolled-unit counts have Pearson correlation 0.94, approved-unit counts 0.90 and grades 0.84. PCA can compact such shared variation, but the resulting components are less interpretable than the original variables.

The trade-off is a 4.78% loss of variance. Explained variance describes the preprocessed inputs, not prediction accuracy or preservation of all target-relevant information. Numerical features enter PCA with variance one, while binary and one-hot indicators keep variance $p(1-p)$, where p is their training prevalence. Features therefore do not have equal variance before PCA; this scaling choice affects both the components and the curve. Models with and without PCA must be compared to assess predictive value.

Prediction timing and scope

The current predictors include results from both semesters. They cannot support a prediction made at enrolment, when these results are unavailable. An early-warning model would need features available at its prediction date and a new preprocessing fit. No predictive model is evaluated in this report.

Conclusion

All 4,424 observations are retained. Encoding, training-based scaling and a stratified split prepare the data for modelling; optional PCA reduces 245 features to 51 components with 95.22% retained variance.

Reference

Realinho, V., Vieira Martins, M., Machado, J., and Baptista, L. (2021). *Predict Students' Dropout and Academic Success*. UCI Machine Learning Repository. DOI: 10.24432/C5MC89.

Data analysed: the course-supplied `graduation_dataset.csv`. Reproducible analysis: `notebooks/student_graduation_assignment.ipynb`.

7. Food Bank supplement: tasks 2 and 3

The supplied crop1.csv adds a case where missing values require treatment. Filtering Element to Production gives 663,735 records across 245 areas, 118 products and 1961-2020. All retained units are tonnes; series keys are complete and unique. Zeros are preserved. live1.csv is not used because the crop subset is sufficient for these tasks.

7a. Missing production values (task 2)

Initially, 56,617 records (8.53%) lack production values. Interpolation is performed within each area-product series, using actual year distances. It assumes a locally linear change and avoids mixing different production scales. Only internal gaps of at most three missing calendar years are filled; no extrapolation or new year records are introduced.

The gap limit is a conservative assumption, not an optimized threshold. Unrestricted interpolation would fill 1,494 values and bridges a 47-year interval in one series. Limiting gaps fills 120; the other 1,374 candidate estimates are rejected. Unsupported values are removed, leaving 607,238 records (91.49% of input). Figure 6 makes this trade-off explicit.

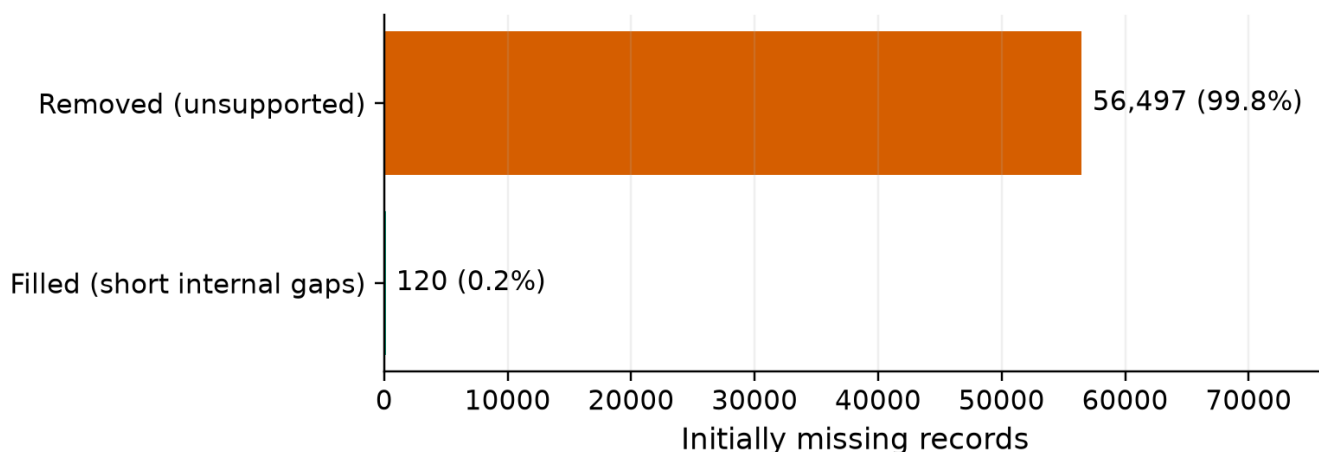


Figure 6. Treatment of initially missing production values. Percentages use the missing-record count as the denominator.

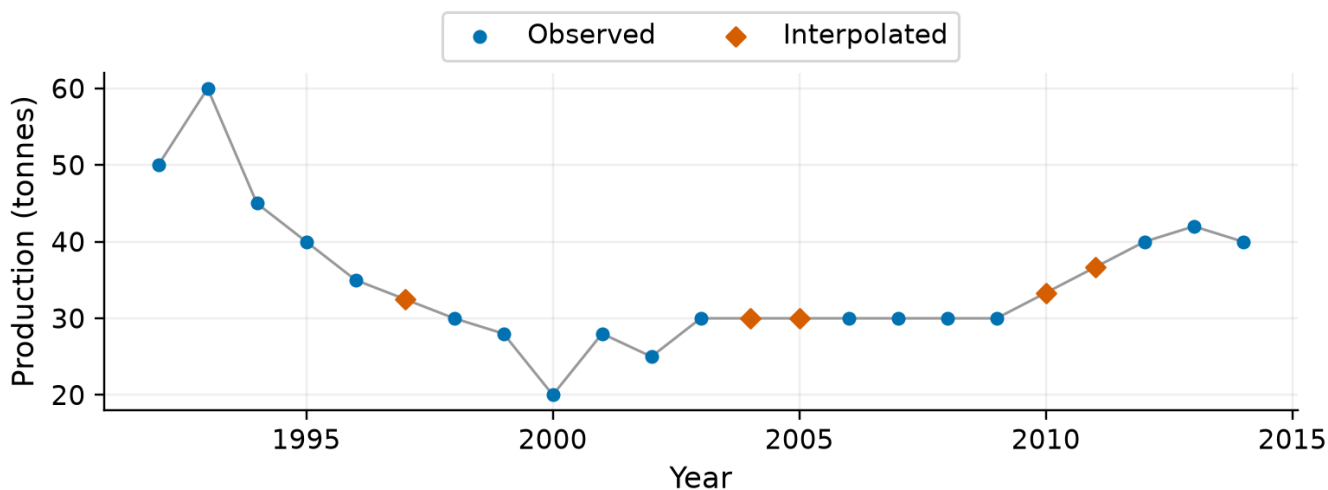


Figure 7. Actual interpolation for Georgia, Anise, badian, fennel, coriander. The series with most accepted estimates is selected for illustration, not as a representative sample.

Removal may disproportionately affect poorly covered areas/products. Short-gap interpolation can still smooth real shocks and uses later observations, so this retrospective method must be reconsidered for forecasting. Original values and a Was_imputed indicator remain available for inspection.

7. Food Bank supplement (continued)

7b. Outlier screening and transformation (task 3)

The pooled 1.5 x IQR rule on retained values gives bounds of -412,879 and 697,380 tonnes and flags 101,942 records (16.79%). Quartiles reduce sensitivity to extreme values, but pooling areas, products and years mixes different production scales. Equal units do not make these observations directly comparable.

The largest flagged records are World sugar-cane totals, with the maximum about 1.96 billion tonnes in 2019. This confirms that regional/global aggregates occur alongside country records; they must not be added together as independent contributions. The pooled flags alone do not demonstrate recording errors. Values are retained rather than removed or capped, pending any source-specific investigation.

A separate $\log_{1p}(\text{Value})$ feature compresses the influence of large magnitudes and supports zero production. The original tonnes remain available. Figure 8 compares the full ranges before and after transformation; no additional observations are deleted.

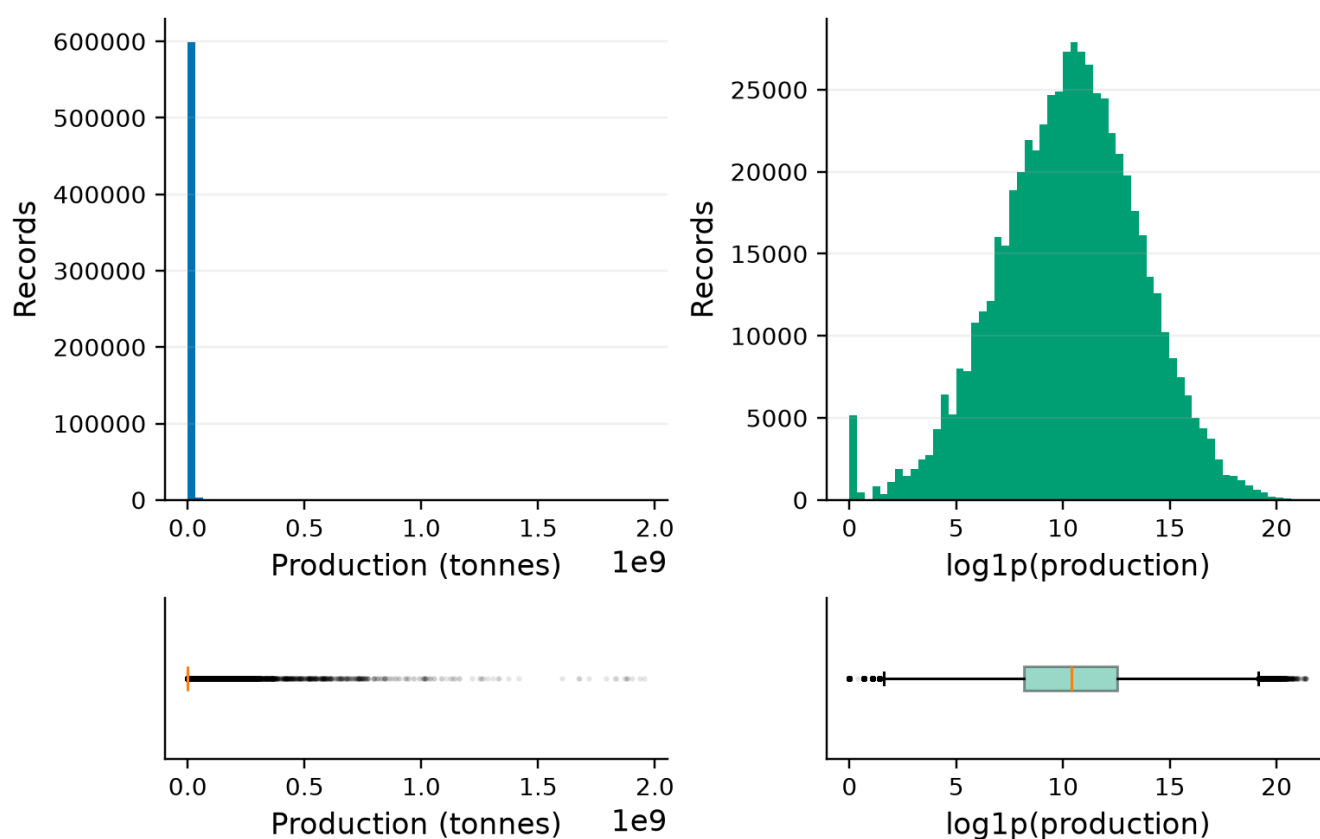


Figure 8. Histograms and box plots before/after $\log_{1p}$, with linear axes in each panel. The full original range concentrates most records near zero. Each box plot computes its whiskers on its own scale, so log-space outliers can differ from the original IQR flags.

Skewness changes from 31.14 to -0.22. This indicates a more symmetric distribution, not proof of normality, error correction or better prediction. All 607,238 retained records survive the transformation. Its predictive value would need to be tested for the intended model.

Supplement outcome

Food Bank provides actual missing-value treatment and a scale-reducing transformation, complementing the justified decision to retain the complete student data. The interpolation limit, excluded rows and original-scale values are documented so the preprocessing can be reviewed.

Source: supplied Food Bank crop1.csv. Reproducible analysis and additional missingness diagnostics: [notebooks/food_bank_preprocessing.ipynb](#).

Appendix A. Column types and roles

Stored datatypes are read from the supplied CSV with pandas. Semantic roles determine preprocessing: an integer code is not automatically a numerical measurement. CU1 / CU2 abbreviate Curricular units 1st / 2nd sem.

Column	Stored dtype	Role
Marital status	int64	Categorical
Application mode	int64	Categorical
Application order	int64	Numerical
Course	int64	Categorical
Daytime/evening attendance	int64	Categorical
Previous qualification	int64	Categorical
Nationality	int64	Categorical
Mother's qualification	int64	Categorical
Father's qualification	int64	Categorical
Mother's occupation	int64	Categorical
Father's occupation	int64	Categorical
Displaced	int64	Categorical
Educational special needs	int64	Categorical
Debtor	int64	Categorical
Tuition fees up to date	int64	Categorical
Gender	int64	Categorical
Scholarship holder	int64	Categorical
Age at enrollment	int64	Numerical
International	int64	Categorical
CU1 (credited)	int64	Numerical
CU1 (enrolled)	int64	Numerical
CU1 (evaluations)	int64	Numerical
CU1 (approved)	int64	Numerical
CU1 (grade)	float64	Numerical
CU1 (without evaluations)	int64	Numerical
CU2 (credited)	int64	Numerical
CU2 (enrolled)	int64	Numerical
CU2 (evaluations)	int64	Numerical
CU2 (approved)	int64	Numerical
CU2 (grade)	float64	Numerical
CU2 (without evaluations)	int64	Numerical
Unemployment rate	float64	Numerical
Inflation rate	float64	Numerical
GDP	float64	Numerical
Target	str	Target

Appendix B. Numerical statistics

All numerical features: descriptive statistics

Every numerical feature has 4,424 non-missing observations. SD is the sample standard deviation; Q1 and Q3 are the 25th and 75th percentiles. Statistics are rounded to two decimals. CU1 / CU2 denote first- / second-semester curricular units.

Feature	Mean	SD	Min	Q1	Median	Q3	Max
Application order	1.73	1.31	0.00	1.00	1.00	2.00	9.00
Age at enrollment	23.27	7.59	17.00	19.00	20.00	25.00	70.00
CU1 (credited)	0.71	2.36	0.00	0.00	0.00	0.00	20.00
CU1 (enrolled)	6.27	2.48	0.00	5.00	6.00	7.00	26.00
CU1 (evaluations)	8.30	4.18	0.00	6.00	8.00	10.00	45.00
CU1 (approved)	4.71	3.09	0.00	3.00	5.00	6.00	26.00
CU1 (grade)	10.64	4.84	0.00	11.00	12.29	13.40	18.88
CU1 (without evaluations)	0.14	0.69	0.00	0.00	0.00	0.00	12.00
CU2 (credited)	0.54	1.92	0.00	0.00	0.00	0.00	19.00
CU2 (enrolled)	6.23	2.20	0.00	5.00	6.00	7.00	23.00
CU2 (evaluations)	8.06	3.95	0.00	6.00	8.00	10.00	33.00
CU2 (approved)	4.44	3.01	0.00	2.00	5.00	6.00	20.00
CU2 (grade)	10.23	5.21	0.00	10.75	12.20	13.33	18.57
CU2 (without evaluations)	0.15	0.75	0.00	0.00	0.00	0.00	12.00
Unemployment rate	11.57	2.66	7.60	9.40	11.10	13.90	16.20
Inflation rate	1.23	1.38	-0.80	0.30	1.40	2.60	3.70
GDP	0.00	2.27	-4.06	-1.70	0.32	1.79	3.51

Categorical cardinalities and observed values are reported in Section 1b. Together with Appendix A, these tables document the datatype and distribution checks behind the preprocessing decisions.